

Teacher: _____

MATHEMATICS

Instructions:

- Write your name and class at the top of this page, and at the top of each answer sheet
- At the end of the examination this examination paper must be attached to the front of your answers.
- All questions are of equal value and may be attempted (12 marks each).
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- A table of standard integrals is attached.

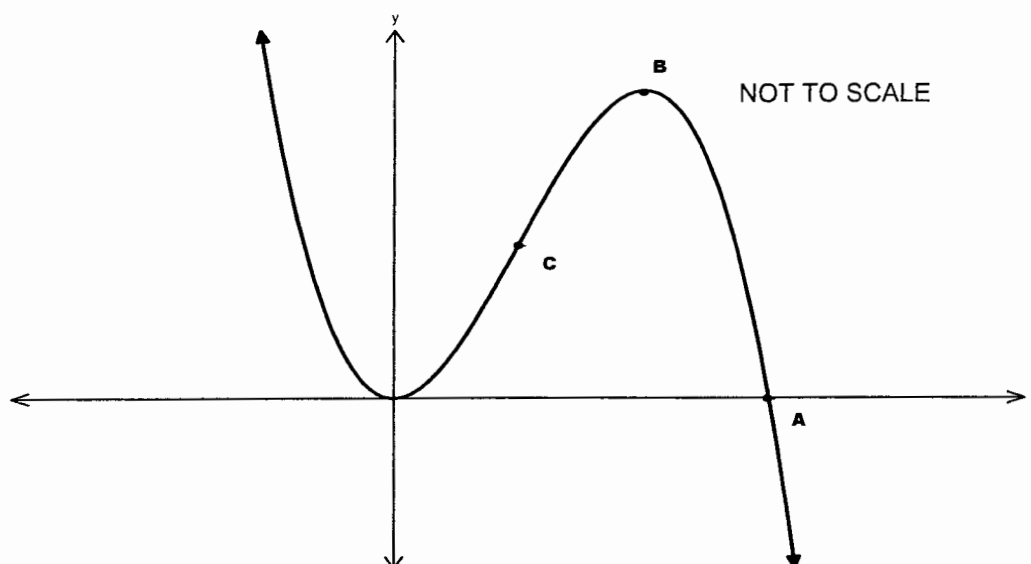
[illegible]

Question 1: (12 marks)**Marks**

- a) Expand and simplify $(\sqrt{2} - 3)^2$ 1
- b) Find $e^{-0.6}$ correct to three decimal places. 2
- c) Find the compound interest earned on \$80 000 if invested for three years at a rate of 6% per annum compounding quarterly. 2
- d) Solve the equation $4x^2 = x$ 2
- e) Solve the equation $|4 - x| = 2x$. 2
- f) Sketch the parabola $x^2 = -4y + 8$ showing its focus and directrix. 3

Question 2: (12 marks) (Start a new page)

a)



The graph represents the function $y = 6x^2 - x^3$.

The point A is an x intercept.

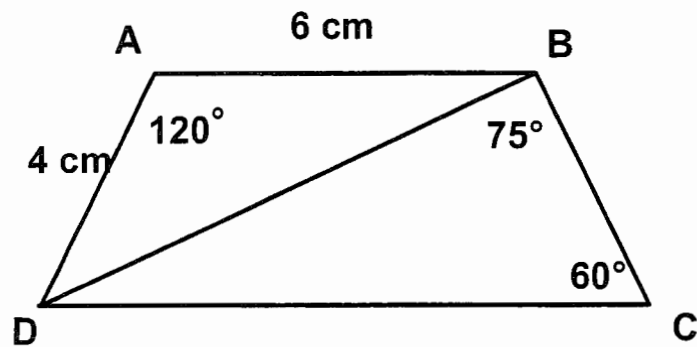
The point B is a local maximum.

The point C is a point of inflexion.

Find

- (i) the coordinates of A 1
- (ii) the coordinates of B 2
- (iii) the coordinates of C 2

b)



- | | | |
|-------|---|---|
| (i) | Find the length of BD as a simplified surd. | 2 |
| (ii) | Find the length of BC correct to one decimal place. | 3 |
| (iii) | Find the area of triangle ABD as a surd. | 2 |

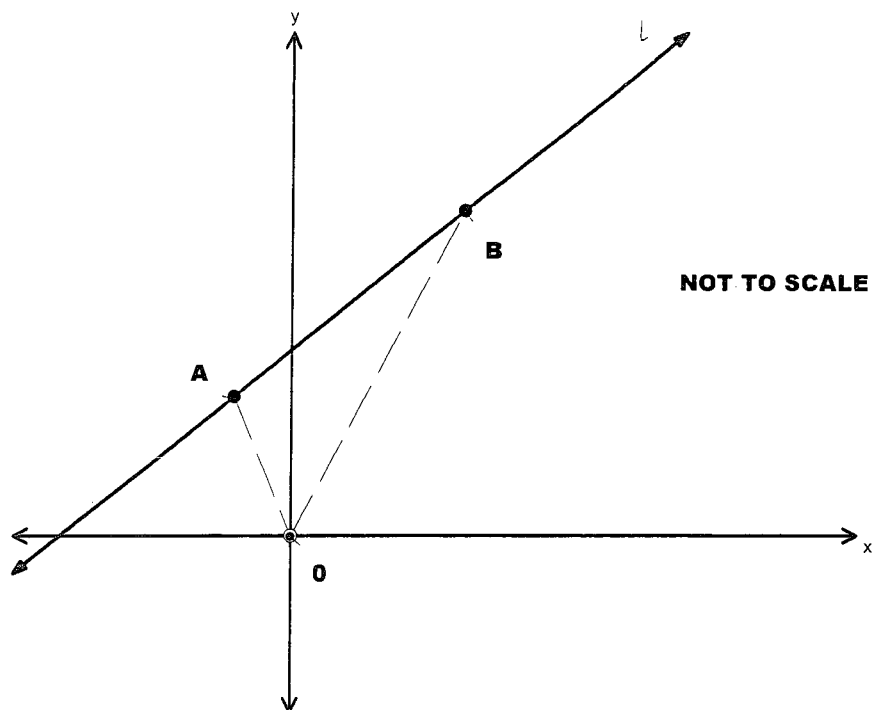
Question 3: (12 marks) (Start a new page)

- | | | |
|------|--|---|
| a) | The first term of an arithmetic sequence is 4 and the fifth term is four times the third term. Find the common difference. | 2 |
| b) | Determine the derivatives of: | |
| (i) | $(3x + 7)^{14}$ | 1 |
| (ii) | $\frac{2x}{x^2-1}$ | 2 |
| c) | Find the EXACT values of the following definite integrals: | |
| (i) | $\int_0^1 e^{3x} dx$ | 2 |
| (ii) | $\int_0^1 \frac{1}{1+x} dx$ | 2 |
| d) | The sum of the first four terms of a geometric sequence is 30 and the limiting sum is 32. If the common ratio is negative, find the common ratio and the first term. | 3 |

Question 4: (12 marks) (Start a new page)

- | | | |
|----|---|---|
| a) | Find the equation of the line (in general form) perpendicular to $2x - 3y - 6 = 0$ and intersecting it on the x axis. | 3 |
|----|---|---|

- b) The line l passes through A $(-1,3)$ and B $(3,7)$.



- | | | |
|-------|--|---|
| (i) | Find the length of AB (in exact form) | 1 |
| (ii) | Find the equation of the line l . | 2 |
| (iii) | Show that the distance from O to the line l is $\frac{4}{\sqrt{2}}$ units. | 2 |
| (iv) | Calculate the area of $\triangle AOB$. | 1 |

- c) The points A $(2,-6)$ and B $(4,8)$ are at opposite ends of the diameter of a circle.

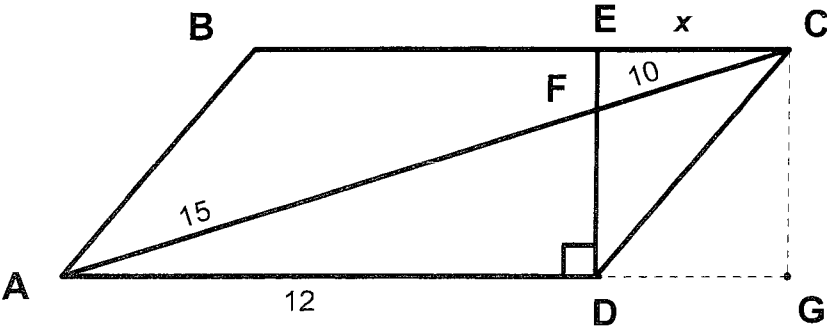
Find

- | | | |
|-------|-----------------------------|---|
| (i) | the centre of the circle. | 1 |
| (ii) | the radius of the circle. | 1 |
| (iii) | the equation of the circle. | 1 |

Question 5:

- a) Given $f'(x) = 3x^2 - 4$,
find $y = f(x)$ if the function passes through (3, 8) 2

b)



$EC = x$
 $AD = 12$
 $FC = 10$
 $AF = 15$
 $ED \parallel CG$

NOT TO SCALE

ABCD is a parallelogram.

Copy the above diagram onto your writing paper

- (i) Prove $\triangle EFC \parallel \triangle DFA$. 2
- (ii) Find the value of x . (with a reason) 2
- (iii) Find the length of CG. (with reasons) 2
- c) A person wishes to invest \$A at the beginning of each month at a compound interest rate of 0.6% per month. How much does the person invest each month in order to have \$20 000 saved at the end of the first year? 4

Question 6: (12 marks) (Start a new page)

- a) (i) Sketch $y = x^2 + 6$ and $y = 12 - x$ on the same axes. 3
Find the x coordinate of the points in intersection.
- (ii) Find the area in the first quadrant bounded by the y axis, $y = x^2 + 6$ and $y = 12 - x$. 2
- b) Use Simpson's Rule with 5 function values to estimate 3
 $\int_0^4 \sqrt{5 + x^2} dx$ correct to 2 decimal places.
- c) Find $\log_{11} 57$ correct to 2 decimal places. 1
- d) (i) Use logarithm laws to simplify $\ln\left(\frac{\sqrt{x-1}}{x^2+1}\right)$ 1
- (ii) Hence find $\frac{d}{dx}\left(\ln \frac{\sqrt{x-1}}{x^2+1}\right)$ 2

Question 7: (12 marks) (Start a new page)

- a) The size of a colony of insects is given by the equation
 $P = 3000e^{kt}$
Where P is the population after t days.
- (i) Write down the initial population. 1
- (ii) If there are 3600 insects after one day, find the value of k , correct to 2 decimal places. 2
- (iii) When will the colony double its initial population? (Answer correct to the nearest day). 2
- (iv) What is the rate of growth of the population after 2 days? 2
- b) Find the volume of the solid formed when the curve $y = \sqrt{x}$ is rotated about the x axis between $x = 1$ and $x = 5$. (Leave your answer in terms of π) 2

- c) Consider the equation $2x^2 - (3 + k)x + 2 = 0$.

For what values of k does the equation have

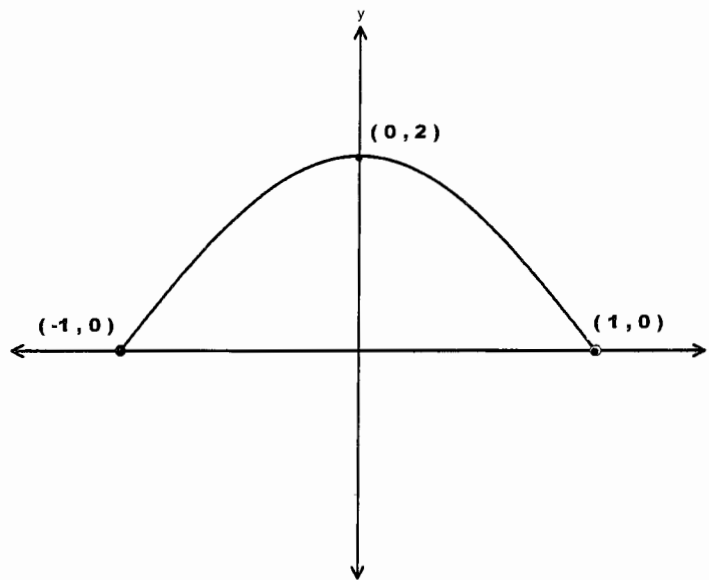
- | | | |
|------|----------------------|---|
| (i) | equal roots | 2 |
| (ii) | different real roots | 1 |

Question 8: (12 marks) (Start a new page)

- a) Differentiate

- | | | |
|-------|------------------|---|
| (i) | $\sin(1 - 2x^3)$ | 2 |
| (ii) | $\tan 3x$ | 2 |
| (iii) | $\cos^2 x$ | 2 |

- b)



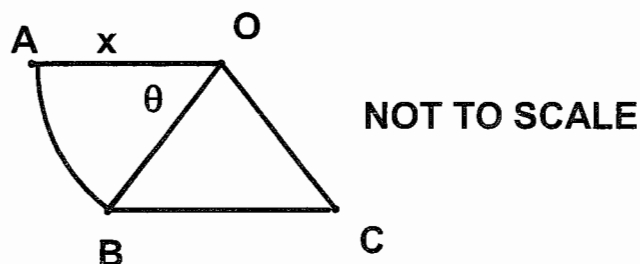
An ornamental arch window 2 metres wide and 2 metres high is to be made in the shape of an arc of either a cosine curve or a parabola, as illustrated on axes above.

- | | | |
|-----|--|---|
| i) | If the arch is made in the shape of the curve $y = 2 \cos \frac{\pi}{2} x$,
find the area of the window. (Your answer may be left in terms of π). | 2 |
| ii) | If the arch is made the shape of an arc of a parabola, find : | |
| | α) the equation of the parabola | 2 |
| | β) the area of the window | 2 |

Question 9 (12 marks) (Start a new page)

- a) (i) On the same diagram sketch the curve $y = \sin \pi x$ and the line $y = x$, in the domain $-1 \leq x \leq 1$. 3
- (ii) Hence find the number of solutions to the equation $\sin \pi x - x = 0$ in the domain $-1 \leq x \leq 1$. 1

b)



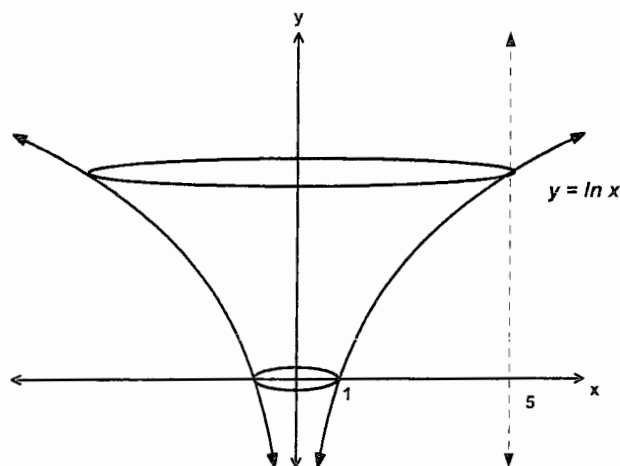
The diagram shows a sector OAB of a circle, centre O and radius x metres. Arc AB subtends an angle θ radians at O. An equilateral triangle BCO adjoins the sector.

Write down expressions for:

- (i) the perimeter of the figure ABCO. 1
- (ii) the area of the figure ABCO. 2
- c) (i) Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$ 2
- (ii) Hence find the equation of the normal to $y = \sec x$ at the point where $x = \frac{\pi}{4}$ (leave your answer in exact form) 3

Question 10 (12 marks) (Start a new page)

a)



NOT TO SCALE

The interior of a bowl is shaped by rotating the arc of the curve $y = \log_e x$ from $x = 1$ to $x = 5$ around the y axis. Calculate the capacity of the bowl in terms of π .

3

b) Given that $a^2 + b^2 = 7ab$, use this result to show that

(i) $\left(\frac{a+b}{3}\right)^2 = ab$

1

(ii) and hence using part i) write

$$\log\left(\frac{a+b}{3}\right) - \frac{1}{2}(\log a + \log b)$$

2

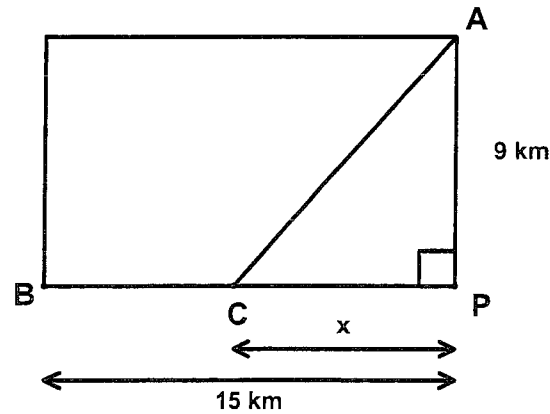
in simplest form.

- c) The diagram shows a rectangular field measuring 9 km by 15 km. From A, a bike rider wishes to go to B.

Riding across the field from A to C, he can average 5km/hr.

Along the road BC, he can average 13 km/hr.

Let $PC = x$



- (i) Show that the time he takes to go from A to C is $\frac{\sqrt{81+x^2}}{5}$ 1
- (ii) Show that the total time he will take to go from A to C to B will be
- $$T = \frac{\sqrt{81+x^2}}{5} + \frac{15-x}{13}$$
- 1
- (iii) Show that the shortest time for the journey will occur when he rides to a point $3\frac{3}{4} \text{ km}$ from P. 4

QUESTION 1

$$1) (\sqrt{2}-3)^2 = 2 - 6\sqrt{2} + 9$$

$$= 11 - 6\sqrt{2}$$

$$2) -0.6$$

$$e = 0.549$$

$$3) \text{ Amt} = 80000(1 + \frac{3}{200})^{12}$$

$$\text{Amt} = \$95649.45$$

$$\therefore \text{Interest} = \$15649.45$$

$$4) 4x^2 = x$$

$$4x^2 - x = 0$$

$$x(4x - 1) = 0$$

$$\therefore x = 0, \frac{1}{4}$$

$$5) |4 - x| = 2x$$

$$4 - x = 2x \quad 4 - x = -2x$$

$$4 = 3x \quad 4 = -x$$

$$x = \frac{4}{3} \quad x = -4$$

check solutions:

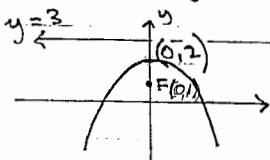
$$|4 - \frac{4}{3}| = 2 \times \frac{4}{3} \quad |4 - (-4)| = -8$$

$$2\frac{2}{3} = 2\frac{2}{3} \quad 8 \neq -8$$

$$\therefore x = \frac{4}{3} \text{ only solution}$$

$$6) x^2 = -4y + 8 \quad \text{Vertex } (0, 2)$$

$$x^2 = -4(y - 2) \quad \text{Focal length } a = 1$$

$$y = 3$$


$$\text{Focus } (0, 1)$$

$$\text{Directrix } y = 3$$

QUESTION 2

$$1) y = 6x^2 - x$$

$$i) y = 0 \quad x^2(6 - x) = 0$$

$$\therefore x = 0, 6$$

$$\therefore A(6, 0)$$

$$ii) \frac{dy}{dx} = 12x - 3x^2$$

$$\text{at pts } y' = 0$$

$$3x(4 - x) = 0$$

$$x = 0 \quad x = 4$$

$$\therefore B(4, 32)$$

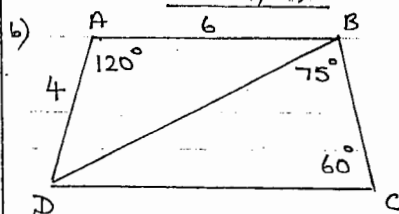
$$iii) \frac{d^2y}{dx^2} = 12 - 6x \text{ put in } y'' = 0$$

$$\frac{d^2y}{dx^2} = 12 - 6x = 0$$

$$12 = 6x$$

$$x = 2$$

$$\therefore C(2, 16)$$



$$i) BD^2 = 6^2 + 4^2 - 2 \cdot 6 \cdot 4 \cdot \cos 120^\circ$$

$$= 36 + 16 - 48x - \frac{1}{2}$$

$$= 52 + 24$$

$$= 76$$

$$\therefore BD = \sqrt{76} \text{ units}$$

$$BD = 2\sqrt{19} \text{ simplified surd}$$

$$ii) \frac{BC}{\sin 45^\circ} = \frac{2\sqrt{19}}{\sin 60^\circ}$$

$$BC = \frac{2\sqrt{19} \cdot \sin 45^\circ}{\sin 60^\circ}$$

$$BC = 7.1 \text{ units}$$

$$iii) \Delta ABD = \frac{1}{2} \cdot 6 \cdot 4 \cdot \sin 120^\circ$$

$$= 12 \cdot \frac{\sqrt{3}}{2}$$

$$= 6\sqrt{3} \text{ units}^2$$

QUESTION 3

$$a) AP: a = 4 \quad T_5 = 4T_3$$

$$a + 4d = 4(a + 2d)$$

$$4 + 4d = 4(4 + 2d)$$

$$4 + 4d = 16 + 8d$$

$$-12 = 4d$$

$$d = -3$$

$$b) i) \frac{d}{dx} (3x+7)^{14} = 14 \cdot 3 (3x+7)^{13}$$

$$= 42 (3x+7)^{13}$$

$$ii) \frac{d}{dx} \left(\frac{2x}{x^2-1} \right) \Rightarrow \text{quotient rule}$$

$$u = 2x \quad v = x^2 - 1$$

$$u' = 2 \quad v' = 2x$$

$$\therefore \frac{d}{dx} \left(\frac{2x}{x^2-1} \right) = \frac{2(x^2-1) - 2x \cdot 2x}{(x^2-1)^2}$$

$$= \frac{2x^2 - 2 - 4x^2}{(x^2-1)^2}$$

$$= \frac{-2x^2 - 2}{(x^2-1)^2}$$

$$c) i) \int_0^1 e^{3x} dx = \left[\frac{e^{3x}}{3} \right]_0^1$$

$$= \frac{1}{3} [e^3 - e^0]$$

$$= \frac{1}{3} (e^3 - 1)$$

$$ii) \int_0^1 \frac{1}{1+x} dx = [\ln(1+x)]_0^1$$

$$= \ln 2 - \ln 1$$

$$= \ln 2$$

$$d) AP: S_4 = 30 \quad S_\infty = 32$$

$$30 = \frac{a(1-r^4)}{1-r} \quad 32 = \frac{a}{1-r}$$

$$\textcircled{1} \div \textcircled{2} \quad \frac{a(1-r^4)}{1-r} \div \frac{a}{1-r} = \frac{30}{32}$$

$$\frac{a(1-r^4)}{1-r} \times \frac{1-r}{a} = \frac{15}{16}$$

$$\frac{1-r^4}{1} = \frac{15}{16}$$

$$16(1-r^4) = 15$$

$$16 - 16r^4 = 15$$

$$1 = 16r^4$$

$$r^4 = \frac{1}{16}$$

$$r = \pm \frac{1}{2} \text{ since negative}$$

$$\therefore \text{from } \textcircled{2} \quad 32 = \frac{a}{1 - (-\frac{1}{2})}$$

$$a = 48$$

QUESTION 4

$$a) \text{ grad of } 2x - 3y - 6 = 0$$

$$3y = 2x - 6$$

$$y = \frac{2x}{3} - 2$$

$$\therefore m_1 = \frac{2}{3} \therefore \text{perp } m_2 = -\frac{3}{2}$$

$$2x - 3y - 6 = 0 \text{ cuts } x \text{ axis}$$

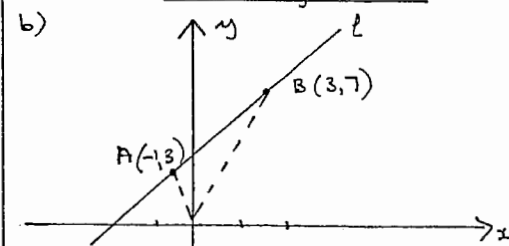
$$\text{at } (3, 0)$$

$$\therefore \text{required line } m_2 = -\frac{3}{2} \text{ thru } (3, 0)$$

$$y - 0 = -\frac{3}{2}(x - 3)$$

$$2y = -3x + 9$$

$$3x + 2y - 9 = 0$$



$$i) AB = \sqrt{(3 - (-1))^2 + (7 - 3)^2}$$

$$= \sqrt{16 + 16}$$

$$= \sqrt{32}$$

$$= 4\sqrt{2} \text{ units}$$

$$ii) l: m = \frac{4}{4} = 1$$

$$y - 7 = 1(x - 3)$$

$$y - 7 = x - 3$$

$$y = x + 4$$

$$iv) \text{ Area } \Delta AOB$$

$$= \frac{1}{2} \cdot 4\sqrt{2} \cdot \frac{4}{\sqrt{2}}$$

$$= 8 \text{ units}^2$$

$$\text{General form}$$

$$iii) p = \left| \frac{0 \cdot 1 + 0 \cdot (-1) + 4}{\sqrt{1^2 + 1^2}} \right|$$

$$p = \left| \frac{4}{\sqrt{2}} \right| \therefore \text{perp dist is } \frac{4}{\sqrt{2}} \text{ units}$$

c) i) Centre (3, 1)

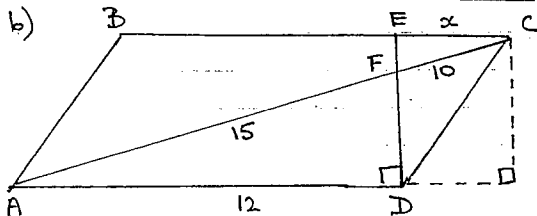
ii) $R = \sqrt{(3-4)^2 + (1-8)^2}$
 $= \sqrt{1^2 + 7^2}$
 $= \sqrt{50}$
 $= 5\sqrt{2} \text{ units}$

iii) $(x-3)^2 + (y-1)^2 = 50$

QUESTION 5

a) $f'(x) = 3x^2 - 4$
 $f(x) = x^3 - 4x + c$
 sub (3, 8) $8 = 27 - 12 + c$
 $8 = 15 + c$
 $c = -7$

$\therefore f(x) = x^3 - 4x - 7$



i) In Δ 's EFC, DFA
 $\widehat{EFC} = \widehat{DFA}$ (vertically opposite)
 $\widehat{ECF} = \widehat{FAD}$ (alternate $BC \parallel AD$)
 $\therefore \Delta EFC \parallel \Delta DFA$ (equiangular)

ii) $\frac{DC}{12} = \frac{10}{15}$ (ratio of corresp. sides in similar triangles)
 $15x = 120$
 $x = 8$

iii) Pythag theorem in $\Delta AFD \therefore FD = 9$
 " " in $\Delta ECF \therefore EF = 6$
 $\therefore ED = 15$
 $ED = CA = 15$ (opp sides rectangle)

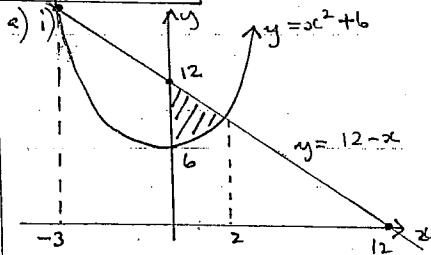
c) $20,000 = A(1.006)^{12} + A(1.006)^{11} + \dots + A(1.006)^1$
 $20,000 = A[1.006 + \dots + 1.006^{12}]$
 $AP: a = 1.006 \quad r = 1.006$
 $n = 12$

$20,000 = A \left[\frac{1.006(1.006^{12}-1)}{1.006-1} \right]$

$\frac{20,000(1.006)}{1.006(1.006^{12}-1)} = A$

$\therefore A = \$1602.76$

QUESTION 6



$x^2 + 6 = 12 - x$
 $x^2 + x - 6 = 0$
 $(x+3)(x-2) = 0$
 $x = -3 \quad x = 2$

ii) $A = \int_{-3}^2 (12-x) - (x^2+6) dx$
 $A = \int_{-3}^2 (6-x-x^2) dx$
 $= \left[6x - \frac{x^2}{2} - \frac{x^3}{3} \right]_{-3}^2$
 $= \left(12 - 2 - \frac{8}{3} \right) - 0$
 $= 7\frac{1}{3} \text{ units}^2$

b)

x	0	1	2	3	4
y	5	6	3	14	21
	F	y ₁	y ₂	y ₃	L

$4 \int_0^4 \sqrt{5+x^2} dx = \frac{1}{3} [15 + 12] + 4(16 + 14)$
 $+ 2.3$
 $= 12.53$

c) $\log_{11} 57 = \frac{\ln 57}{\ln 11}$
 $= 1.69$

d) i) $\ln \left(\frac{\sqrt{x-1}}{x^2+1} \right) = \ln(\sqrt{x-1}) - \ln(x^2+1)$
 $= \frac{1}{2} \ln(x-1) - \ln(x^2+1)$

ii) $\frac{d}{dx} \left(\frac{\sqrt{x-1}}{x^2+1} \right) = \frac{d}{dx} \left(\frac{1}{2} \ln(x-1) - \ln(x^2+1) \right)$
 $= \frac{1}{2} \frac{1}{(x-1)} - \frac{2x}{x^2+1}$
 $= \frac{1}{2(x-1)} - \frac{2x}{x^2+1}$

QUESTION 7

a) $P = 3000e^{-kt}$

i) $t = 0 \quad P = 3000$

ii) $t = 1 \quad P = 3600$

$3600 = 3000e^{-k}$
 $\frac{36}{30} = e^{-k}$

$\ln \frac{6}{5} = -k$

$k = 0.18 \quad (2 \text{ dec pl})$

iii) $6000 = 3000e^{.18t}$
 $2 = e^{.18t}$
 $\ln 2 = .18t$
 $\therefore t = \frac{\ln 2}{.18}$

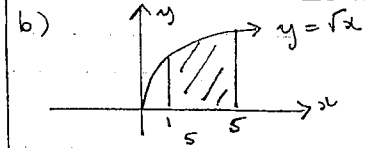
$t = 3.85 \therefore t = 4 \text{ days}$

iv) $\frac{dP}{dt} = -k \cdot 3000e^{-kt}$

$t = 2$

$\frac{dP}{dt} = .18 \times 3000 \times e^{-.18 \times 2}$

$= 774 \text{ insects/day}$



$V_x = \pi \int_0^5 (\sqrt{x})^2 dx$
 $= \pi \int_0^5 x dx$
 $= \pi \left[\frac{x^2}{2} \right]_0^5$
 $= \frac{\pi}{2} [25 - 0]$
 $= 12\pi \text{ units}^3$

c) i) $\Delta = 0$
 $(3+k)^2 - 4 \cdot 2 \cdot 2 = 0$
 $9 + 6k + k^2 - 16 = 0$
 $k^2 + 6k - 7 = 0$
 $(k+7)(k-1) = 0$
 $k = -7 \quad k = 1$

ii) $\Delta > 0$
 $k > 1, k < -7$

QUESTION 8

a) i) $\frac{d}{dx} (\sin(1-2x^3)) =$

$-6x^2 \cos(1-2x^3)$

ii) $\frac{d}{dx} (\tan 3x)$
 $= 3 \sec^2 3x$

iii) $\frac{d}{dx} (\cos x)^2$
 $= -2 \cdot \sin x \cdot \cos x$

b) i) $A = 2 \int_0^1 2 \cos \frac{\pi}{2} x dx$
 $= 2 \left[\frac{2.2}{\pi} \sin \frac{\pi}{2} x \right]_0^1$

$$= \frac{8}{\pi} \left[\sin \frac{\pi}{2} x \right]_0^1$$

$$= \frac{8}{\pi} \left[\sin \frac{\pi}{2} - \sin 0 \right]$$

$$= \frac{8}{\pi} \text{ unit}^2$$

ii) a) use $(x-h)^2 = -4a(y-k)$
vertex $(0, 2)$
 $x^2 = -4a(y-2)$
sub pt $(1, 0)$
 $1 = -4a \cdot -2$
 $1 = 8a$
 $a = 1/8$

eqn parab: $x^2 = -\frac{1}{2}(y-2)$

b) $2x^2 = -y + 2$
 $y = 2 - 2x^2$

$$A = 2 \int_0^1 (2 - 2x^2) dx$$

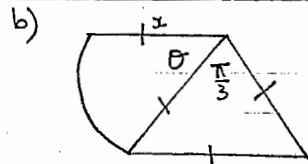
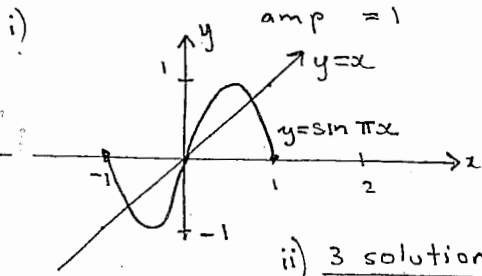
$$= 2 \left[2x - \frac{2x^3}{3} \right]_0^1$$

$$= 2 \left[2 - \frac{2}{3} \right]$$

$$= \frac{8}{3} \text{ unit}^2$$

QUESTION 9

a) $y = \sin \pi x$ period = $\frac{2\pi}{\pi} = 2$



i) $P = 3x + x\theta$

ii) $A = \frac{1}{2}x^2\theta + \frac{1}{2}x^2\sin\frac{\pi}{3}$
 $= \frac{1}{2}x^2\theta + \frac{1}{2}x^2\frac{\sqrt{3}}{2}$
 $= \frac{1}{2}x^2\theta + \frac{\sqrt{3}}{4}x^2$

c) $\frac{d}{dx}(\sec x) = \frac{d}{dx}(\cos x)^{-1}$

i) $\frac{d}{dx} = -1 \cdot \sin x (\cos x)^{-2}$
 $= \frac{\sin x}{\cos^2 x}$
 $= \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x}$
 $= \tan x \cdot \sec x$

ii) $m_T = \tan \frac{\pi}{4} \cdot \sec \frac{\pi}{4}$

$m_T = 1 \cdot \sqrt{2}$

$m_N = -\frac{1}{\sqrt{2}}$

at $x = \frac{\pi}{4}$ $y = \sec \frac{\pi}{4}$

$y = \sqrt{2}$

pt $(\frac{\pi}{4}, \sqrt{2})$ and $m_N = -\frac{1}{\sqrt{2}}$

normal $y - \sqrt{2} = -\frac{1}{\sqrt{2}}(x - \frac{\pi}{4})$

$\sqrt{2}y - 2 = -x + \frac{\pi}{4}$

$x + \sqrt{2}y - 2 - \frac{\pi}{4} = 0$

QUESTION 10

a) $x=1 \Rightarrow y=0$ $x=5 \Rightarrow y=1.5$
 $V = \pi \int_0^{1.5} (e^y)^2 dy$ $y = \log e^{2x}$
 $e = x$

$V = \pi \left[\frac{e^{2y}}{2} \right]_0^{1.5}$

$V = \frac{\pi}{2} [e^{2 \ln 5} - e^0]$

$= \frac{\pi}{2} [e^{2 \cdot 5} - 1]$

$= \frac{24\pi}{2}$

$= 12\pi \text{ unit}^3$

b) i) $\left(\frac{a+b}{3}\right)^2 = \frac{a^2 + 2ab + b^2}{9}$
 $= \frac{2ab + 7ab}{9}$

$= ab$

$LHS = RHS$

ii) $\log \left(\frac{a+b}{3}\right)^2 = \log ab$

$2 \log \left(\frac{a+b}{3}\right) = \log ab$

$\log \left(\frac{a+b}{3}\right) = \frac{1}{2} \log ab$

$\log \left(\frac{a+b}{3}\right) = \frac{1}{2} (\log a + \log b)$

$\therefore \log \left(\frac{a+b}{3}\right) - \frac{1}{2} [\log a + \log b] = 0$

c) i) $AC = \sqrt{x^2 + 81}$ pyth. th.

$D = S \cdot T \therefore T = \frac{D}{S}$

$\therefore T_{AC} = \frac{\sqrt{x^2 + 81}}{5}$

ii) $T_{BC} = \frac{15-x}{13}$

$\therefore \text{Total Time}$

$T = \frac{\sqrt{x^2 + 81}}{5} + \frac{15-x}{13}$

iii) $\frac{dT}{dx} = \frac{1}{2} \cdot \frac{2x}{5} (x^2 + 81)^{-\frac{1}{2}} - \frac{1}{13}$

$\frac{dT}{dx} = \frac{x}{5\sqrt{x^2 + 81}} - \frac{1}{13}$

set pt $T' = 0$

$\frac{x}{5\sqrt{x^2 + 81}} = \frac{1}{13}$

$13x = 5\sqrt{x^2 + 81}$

$169x^2 = 25(x^2 + 81)$

$169x^2 = 25x^2 + 2025$

$144x^2 = 2025$

$x^2 = \frac{2025}{144}$

$x = \pm \frac{45}{12}$

$x > 0$

test max/min for $x = 3.75$

x	3	$3\frac{3}{4}$	4
T'	-	0	+

$\searrow$ $\swarrow$
- $\frac{0}{+}$

$\therefore \text{min Time if } x = 3.75$